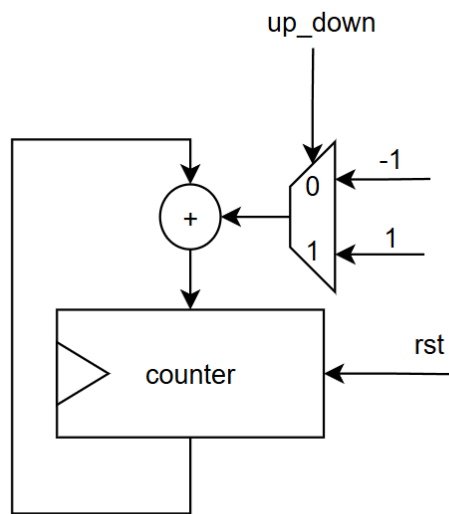


## Task 10

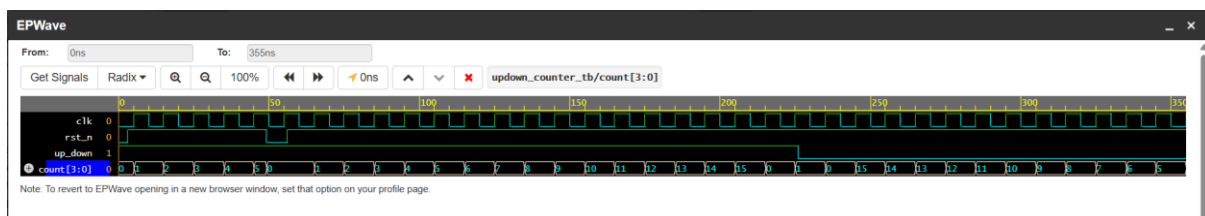
<https://edaplayground.com/x/KFWF>



*4-bit Updown Counter*

```
Log Share
CPU time: .577 seconds to compile + .528 seconds to elab + .526 seconds to link
Chronologic VCS simulator copyright 1991-2025
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64; Jul 20 11:37 2026
PASS: Asynchronous reset works.
PASS: count=1
PASS: count=14
$finish called from file "testbench.sv", line 85.
$finish at simulation time 346
VCS Simulation Report
Time: 346 ns
CPU Time: 0.480 seconds; Data structure size: 0.0Mb
Mon Jul 20 11:37:21 2026
Finding VCD file...
./counter.vcd
[2026-07-20 15:37:22 UTC] Opening EPWave...
Done
```

*Terminal Output Screenshot*



*Waveform Screenshot*